THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which one of these pairs contains only physical changes? 1

(a) burning wood, rusting **(b)** digestion, photosynthesis
(c) freezing of water, evaporation of water **(d)** cooking, curdling

2. In $H_2SO_4 + 2\,NaOH \longrightarrow Na_2SO_4 + 2\,H_2O$, the salt formed is: 1

(a) sodium sulphate **(b)** sodium chloride **(c)** sodium nitrate **(d)** sodium carbonate

3. An electrician wears rubber gloves because they: 1

(a) are costly **(b)** are insulators **(c)** are heavy **(d)** are conductors

4. When sulphur is burnt and gas passed into water, the solution turns: 1

(a) no change **(b)** blue litmus red **(c)** red litmus blue **(d)** green

5. In the experiment, the nail in boiled water with oil does not rust because: 1

(a) no water **(b)** too much oil
(c) no air reaches it **(d)** no iron

6. Why does baking soda taste bitter? 1

(a) it is a base **(b)** it is a salt **(c)** it is sugar **(d)** it is acid

7. A student wants to test an object for conduction. Which item completes the circuit tester? 1

(a) a cell, bulb and wires **(b)** only a switch
(c) only a bulb **(d)** only wires

8. Why is silver used in mirrors and in jewellery? 1

(a) it is dull **(b)** it is gaseous **(c)** it is brittle **(d)** it is lustrous

9. In making crystals, why is the solution filtered while hot? 1

(a) to make it solid **(b)** to make it blue
(c) to remove all water **(d)** to remove insoluble impurities

10. A solution does not change the colour of litmus or turmeric. It is: 1

(a) acidic **(b)** basic **(c)** neutral **(d)** salty only

11. A bulb does not glow in a circuit although the cell is new. The most likely reason is: 1

(a) the filament is broken or the circuit is open
(b) the wires are copper
(c) the cell is full
(d) the switch is closed

12. Which of these is a metal that is also a poor conductor of heat compared with others? 1

(a) lead **(b)** aluminium **(c)** silver **(d)** copper

13. Which of the following changes cannot be reversed? 1

(a) freezing of water **(b)** frying of an egg
(c) folding of paper **(d)** melting of butter

14. In $\mathrm{HNO_3 + KOH \longrightarrow KNO_3 + H_2O}$, the salt formed is: 1

(a) potassium sulphate **(b)** potassium chloride
(c) sodium nitrate **(d)** potassium nitrate

15. Which arrangement correctly connects a cell and bulb? 1

(a) only one terminal connected
(b) both bulb terminals to the same cell terminal
(c) each bulb terminal connected to a different cell terminal
(d) no wires

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are cells not connected to opposite polarity in series? 2

17. Give two examples where metal oxides are used as bases. 2

18. Why are crystals formed when a hot saturated solution is cooled? 2

19. Differentiate between a salt and an acid. 2

20. What precaution should you take while replacing a fuse? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Why is photosynthesis a chemical change? 3

22. Why should chemicals in the lab be labelled? 3

23. Describe a safe way to disconnect a plugged-in device. 3

24. How do metals help in conduction of heat in cooking? 3

OR

Compare cooking of rice with melting of butter. 3

25. Describe how to prepare a litmus-like indicator from beetroot or red cabbage and test with vinegar and soap. 3

OR

Describe how you would design a simple circuit tester and test five objects. 3

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SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A student mixes 10 mL of vinegar with baking soda. Describe the observation and explain the reaction in words. 5

OR

Describe how an electrician stays safe while repairing a socket. 5

27. Describe a safe procedure to burn magnesium and test the product. 5

OR

Discuss the harmful and useful chemical changes with three examples each. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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